

Suitability and Reliability of the Dataset

The clinic directory used in the CareQueue system was seeded using a real South African healthcare facility dataset obtained from a publicly available and reliable source called “Humanitarian Data Exchange” . The dataset includes authentic clinic and healthcare facility information such as clinic names, physical locations, provinces, districts, facility types, and regional classifications. This makes the dataset suitable for a healthcare queue management application because it reflects actual healthcare infrastructure available to South African citizens.

The dataset is appropriate for the project because it supports the system requirements for searching and filtering clinics by province, district, region, facility type, and services offered. Using real healthcare data improves the realism and usability of the application compared to using manually generated or fictional clinic records. It also allows users to search for facilities in meaningful geographic areas and provides a more accurate simulation of a real healthcare environment.

The chosen source is considered reliable because it originates from an official government or public health data repository that maintains healthcare facility records for administrative and public service purposes. Government and public-sector datasets are generally trustworthy because they are regularly updated, standardized, and used by healthcare authorities and organizations for planning and reporting. Using an authoritative source increases the credibility and integrity of the project data.

Additionally, the dataset structure integrates well with Firebase Firestore, allowing clinic records to be stored and queried efficiently within the CareQueue system. This supports scalable filtering and searching functionality required by the application.